: Anita Tomar
: Professor and Head
Department of Mathematics

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:
: **Educational Qualifications** : M.Sc. Mathematics (Ist division), 1985
: M.Phil. Mathematics (Ist division), 1986
: Ph. D. in Mathematics- "Coincidences and Fixed Points of Single Valued and Multivalued Maps", 2003

No. of Publications : **104 +14** Chapters in edited books+**3 edited books**
: .
: **Life membership** : Indian Society for History of Mathematics (Membership number- L248)
: Indian Mathematical Society(Membership number- T-01-036)
: Vijñāna Parishad of India: Orai, Uttar Pradesh(422)

: **Area of specialization:** Fixed Point Theory and its Applications, Fractals, Nonlinear Functional Analysis

Awards:

1. **"Best Project Award"** in Advanced Refresher Course (Mathematics), *Indian Institute of Science, Challakere Campus, Khudapura*, August 15–30, 2025.
2. **"Distinguished Service Award",** Vijñāna Parishad of India, (Society for Applications of Mathematics)for outstanding contribution to Subject and/or Vijñāna Parishad of India and/or National Development at the occasion of 26th Annual Conference Cum 7th International Conference of VPI held at SC&IS,JNU, New Delhi, July 28-30, 2025.
3. **"Excellence in Research Award 2023**" by Divya Himgiri, Government of Uttarakhand, Uttarakhand Council of Science and Technology(UCOST), USERC, ONGC, SEED, 4th Dehradun International Science and Technology Festival,6th Himalayan Educator's Summit & Felicitation Ceremony, Oct. 29, 2023.
4. **"Chief Minister Excellence and Good Governance Award 2020-21"** Chief Minister Uttarakhand, Government of Uttarakhand, on the occasion of Good Governance Day, Dec.25, 2022.
5. **"Women Scientist Award**", Uttarakhand Council of Science and Technology (**UCOST),** Dehradun (Uttarakhand), March 08, 2021.
6. "**Teacher of the Year 2020**" by Divya Himgiri, SRADSTA, Uttarakhand Council of Science and Technology(UCOST), Commission for Scientific & Technical Terminology, MHRD, Government of India, Samagra Shiksha Uttarakhand, Uttarakhand Pravidhik Shiksha Parishad, Roorkee (Haridwar), Uttarakhand Technical University, Dehradun, Sridev Suman Uttarakhand University, New Tehri, Sept. 05, 2020.

19. List of Publications:

1. **Anita Tomar**, Swati Antal, Mohammad Sajid, Darshana J. Prajapati. Role of s-convexity in the generation of fractals as Julia and Mandelbrot sets via three-step fixed point iteration, AIMS Mathematics, 2025, 10(11): 26077-26105. https://doi.org/10.3934/math.20251148
2. **Anita Tomar**, Khairul Habib Alam, Mohammad Sajid, Yumnam Rohen, S. Surendra Singh, Fibonacci-Ishikawa iterative method in modular spaces for asymptotically non-expansive monotonic mathematical operators, J. Inequal. Appl. 2025, 126 (2025). *https://doi.org/10.1186/s13660-025-03378-4* .
3. Khairul Habib Alam, Yumnam Rohen, **Anita Tomar**, Fixed Point and its Geometry and Application for Multivalued Integral Type Contractions in $m^b{}_v$−metric spaces, Hacet. J. Math. Stat. *54(05), 1708-1724,* https://doi.org/10.15672/hujms.1471688.
4. Khairul Habib Alam, Yumnam Rohen, **Anita Tomar** and Mohammad Sajid, "Set-valued contractions with an application to Fredholm integral inclusions in mbv−metric spaces, AIMS Mathematics, 2025, 10(9): 20742-20758. doi: *10.3934/math.2025926*
5. Meena Joshi, **Anita Tomar**, Sumaiya Tasneem Zubair, Aiman Mukheimer, Thabet Abdeljawad, Investigation of Near Fixed Points, Near Fixed Interval Ellipse and its Equivalence Classes: On Near Fixed Points and Near Fixed Interval Ellipse .

Results in Nonlinear Analysis, Results in Nonlinear Analysis 8 (2025) No. 2, 284–304 *https://doi.org/10.31838/rna/2025.08.02.022.*

6. Iqbal H. Jebril, Hakima Bouhadjera, **Anita Tomar,** Shivangi Upadhyay, Unique Common Fixed Point for Occasionally Weakly Biased Mappings of Type (A) in Ultrametric Space, Pan-Amer. J. Math. 4 (2025), 11, https://doi.org/10.28919/cpr-pajm/4-11
7. K.H. Alam, Y. Rohen, and **Anita Tomar,** Approximating the solutions of fractional differential equations with a novel and more efficient iteration procedure. J. Super Comput. 81(9), 1084 (2025). https://doi.org/10.1007/s11227-025-07562-7
8. **Anita Tomar,** U. S. Rana, Vipul Kumar, Fixed points of ω-interpolative Hardy-Rogers contraction and its application in b-metric spaces, Afr. Mat., 36(2), 2025,Paper No. 103, 13pp *https://doi.org/10.1007/s13370-025-01320-x*
9. **Anita Tomar,** Saurabh Sharma and Sanjay Kumar Padaliya, Mohammad Sajid, Hybrid Picard S-Iteration for Escape Analysis, Symmetry, Computational Insights, and Visualization of Complex Fractals" Math. Comput. Model. Dyn. Syst., 31(1), 2025, Paper No. 2500444, 23pp. *https://doi.org/10.1080/13873954.2025.2500444*.
10. Jagan Mohan Jonnalagadda, **Anita Tomar**, Positive solutions of discrete fractional Sturm–Liouville problems J. Math. Sci. (2025). *https://doi.org/10.1007/s10958-025-07720-5*

. **20. Chapter in edited Books:**

1. Swati Antal, **Anita Tomar**, and U. C. Gairola, Application of Common Fixed Point in the Framework of b-Metric Space via Simulation Function, Chapter 04. In: Tomar, A., Jain, M. (eds) Banach Contraction Principle. Industrial and Applied Mathematics. Springer, Singapore. 2025, 55-74***(indexed in Scopus)***. **eBook ISBN** 978-981-96-4847-4
2. **Anita Tomar**, Vipul Kumar, and U. S. Rana, Julia Fractals of Complex-Valued Cosine Functions Via Fixed Point Iterations, Chapter 06. In: Tomar, A., Jain, M. (eds) Banach Contraction Principle. Industrial and Applied Mathematics. Springer, Singapore. 2025, 85-117***(indexed in Scopus)***https://doi.org/10.1007/978-981-96-4847-4_6. **eBook ISBN** 978-981-96-4847-4
3. Shivam Rawat, Darshana J. Prajapati, **Anita Tomar**, and R. C. Dimri, Application of Fixed Point Iterations in Generation of Fractals for Higher-Order Complex Polynomial, Chapter 08. In: Tomar, A., Jain, M. (eds) Banach Contraction Principle. Industrial and Applied Mathematics. Springer, Singapore. 2025, 137-161***(indexed in Scopus)***. **eBook ISBN** 978-981-96-4847-4
4. Shivam Rawat, Ayush Bartwal, Monika Bisht and Anita Tomar, "Fixed points and their geometric properties for interpolative contravariant mappings". Nonlinear Analysis and Computational Techniques: Proceedings of the ICNACT-2024 Conference Held During 8-10 August, 2024, edited by Hemant Kumar Nashine, Ranis Ibragimov and Hemanta Kalita, De Gruyter, 2025, pp. 1-14. *https://doi.org/10.1515/9783111724638-001 10.1002/mma.8871 10.1002/mma.8871* **eBook ISBN:**9783111724638, **Hardcover ISBN:**9783111724553***(indexed in Scopus)***.
5. Shivangi Upadhyay and **Anita Tomar,** Values of Trigonometric Functions in Ancient India: A Comparative Analysis with other Countries and their Utility, Chapter 10 in a book Adhunik Pridrishya Me Bharatiya Prachiya Gyaan Sampada (Connecting Ancient Indian Wisdom in Modern Contexts)(Vol . 1), 152-174, Indu Book Services Pvt Ltd, Nov.07, 2024.ISBN : 9789367291658, Hardcover – 7.
6. Meena Joshi and **Anita Tomar,** (2024). Some Generalizations of Fixed Circle. In: Chatzakou, M., Ruzhansky, M., Stoeva, D. (eds) Women in Analysis and PDE. GFOW APDEGS 2022 2021. Trends in Mathematics, vol 5, 221-227. Birkhäuser, Cham. https://doi.org/10.1007/978-3-031-57005-6_23***(indexed in Scopus)***.
7. Shivangi Upadhyay, Meenakshi Rana & **Anita Tomar**, Contribution of Aryabhat in Ancient Mathematics, Chapter in proceedings of the National Conference of Mathematics and its Application in Science Uttarakhand Open University, Haldwani, Nainital, Dec. 22-23, 2022, Pacific Books International, ISBN 978-93-92469-16-9.
8. **Anita Tomar,** Meena Joshi, Dhananjay Kumar, Fermat's Contribution in Mathematics, Chapter in a book, Understanding Values and Ethics in Shrinking World, Bloomsbury Publishing India Pvt. Ltd., March 2022,60-72. ISBN 978-93-54358-94-4.
9. Meena Joshi, **Anita Tomar** and S. K. Padaliya, On geometric properties of non-unique fixed points in b−metric spaces, Chapter in a book, Fixed Point Theory & its Applications to Real World Problem, Nova Science Publishers, New York, USA, 2021, 33-50. (***indexed in Scopus)*** ISBN 978-1-53619-336-7.
10. **Anita Tomar** and Meena Joshi, Near fixed point, near fixed interval circle and near fixed interval disc in metric interval space, Chapter in a book, Fixed Point Theory & its Applications to Real World Problem, Nova Science Publishers, New York, USA,2021, 131-150. (***indexed in Scopus)*** ISBN 978-1-53619-336-7.

21.Ph. D. Degrees Supervised:

1. Ritu Sharma : **Some Fixed Point Theorems and their Applications,** July 23, 2019. Registration No. SM-16150(Apr. 08, 2016).
2. Shivangi Upadhyay: **Existence of Fixed Point and its Applications,** July 24, 2019. Registration No. SM-16152(Apr. 08, 2016).

22.Reviewer:

- Reviewer, Mathematical Reviews a division of American Mathematical Society, USA.
- Reviewer of research papers indexed in MathSciNet, ZbMATH, Scopus, web of science, UGC –CARE list.

23.Editor:

1. **Anita Tomar** and Manish Jain, Banach Contraction Principle: A Centurial Journey. Industrial and Applied Mathematics. Singapore: Springer (ISBN 978-981-964846-7/hbk; 978-981-964849-8/pbk; 978-981-964847-4/ebook). xii, 406 p. (2025). (***indexed in Scopus***).
2. N. K. Joshi, Kalpana, **Anita Tomar**, Poonam Pathak (Author), Adhunik Pridrishya Me Bharatiya Prachiya Gyaan Sampada(Connecting Ancient Indian Wisdom in Modern Contexts)(Set of 2 Vol) (ISBN 9789367291658) Hardcover -7, Indu Book Services Pvt Ltd, Nov.07,2024
3. **Anita Tomar**(**Guest Editor**), Erdal Karapinar, and Juan Martinez-Moreno, Special Issue on Nonlinear Analysis and its applications, *J. Math. Sci.*, 2024, (***indexed in Scopus).*** **eISSN 1573-8795, pISSN1072-3374.**
4. Santosh Kumar, **Anita Tomar**(**Guest Editor**), Cristian Chifu, and Mohammad Mursaleen , Special Issue on "Recent Developments in Fixed-Point Theory and Applications**"**, Demonstr. Math., 2023 (***indexed in Scopus and Web of Science***). ISSN: 2391-4661.
5. **Anita Tomar** (**Lead Editor**), Santosh Kumar and Cristian Chifu, Special Issue on "Unique and Non-Unique Fixed Points and their Applications 2022"**,** J. Funct. Spaces, 2022 (***indexed in Scopus and Web of Science***). ISSN: 2314-8896 (Print) ISSN: 2314-8888 (Online).
6. Santosh Kumar, **Anita Tomar**(**Guest Editor**), Juan Martinez-Moreno**,** Rale M. Nikolić**,** Special Issue on "Fixed-Point Techniques and Applications to Real World Problems", J. Funct. Spaces, 2021 (***Scopus and Web of Science***). ISSN(p): 2314-8896. ISSN(e): 2314-8888.
7. **Anita Tomar** (**Lead Editor**), Santosh Kumar and Liliana Guran, Special Issue on "Unique and Non-Unique Fixed Points and their Applications"**,** J. Funct. Spaces, 2021(***Scopus and Web of Science***). ISSN(p):2314-8896.ISSN(e):2314-8888.
8. **Anita Tomar** and M.C. Joshi, "Fixed Point Theory & its Applications to Real World Problem", Mathematics Research Developments. New York, NY, USA.: Nova Science Publishers (ISBN 978-1-5361-9336-7/hbk; 978-1-5361-9479-1/ebook). x, 412 p. (2021).

24. Authored Books:

1. **Anita Tomar,** Imran Ali and Neetu Ram, '**Fundamental Mathematics-I (Theory of Equations, Matrices and Vector Calculus)'**, Pragati Prakashan, 2025. ISBN: 978-93-5854-304-9 .
2. **Anita Tomar** and Sudheer Petwal, '**Complex Analysis'**, Pragati Prakashan, 2024. ISBN: 978-93-5854-304-9 .
3. P. N. Chatterjee, **Anita Tomar** and Sudheer Petwal, '**Ring Theory'**, Pragati Prakashan, 2023. ISBN:978-93-5854-839-6.
4. **Anita Tomar**, Sudheer Petwal, '**Ring Theory'**, Pragati Prakashan, 2023
5. **Anita Tomar**, Sudheer Petwal **, 'Group Theory',** Pragati Prakashan,2023
6. **Anita Tomar**, Sudheer Petwal, and K.C. Yadav '**Trigonometry'**, Pragati Prakashan, 2022. ISBN:978-93-5531-733-9.
7. **Anita Tomar** and Sudheer Petwal, '**Differential Calculus'**, Pragati Prakashan, 2022. ISBN:978-93-5531-718-6.
8. **Anita Tomar** , Sudheer Petwal, and K.C. Yadav '**Matrices'**, Pragati Prakashan, 2022. ISBN: 978-93-5531730-8.
9. P. N. Chatterjee, Rajneesh Saini, **Anita Tomar** and Himanshu Prajapati, '**Trigonometry'**, Pragati Prakashan, 2018. ISBN:978-93-88151-21-4.
10. P. N. Chatterjee, Mukesh Gupta, Rajneesh Saini, **Anita Tomar** and Imran Ali, '**Integral Calculus',** Pragati Prakashan, 2018. ISBN:978-93-88438-92-6.
11. U. S. Rana, **Anita Tomar** and Ashok Kumar, Book- '**Theory of Equations'**, 2017. Arihant Publications. ISBN: 978-93-86080-37-0.

25. Honours:

1. Attended the Open Editorial Board Meeting of the Journal of Mathematical Sciences (Springer), Georgian National Academy of Sciences, Rustaveli Ave, Tbilisi, Oct. 9-10, 2025.
2. Selected under the Chief Minister's Excellent Research Paper Publication Incentive Scheme 2024, April 19,2025. Letter No.:442/53(mu.u.sho.p.pra. pro. yo.)/2024-25).(10 papers-14751).
3. Attended the Annual Editorial Board Meeting of the Journal of Mathematical Sciences (Springer) , hosted by the University of Vienna, led by Prof. Radu Ioan Boţ (Dean, Faculty of Mathematics, University of Vienna) and Dr. Alexey Karapetyants (Editor-in-Chief, JMS Series A) Nov. 25, 2024.
4. Chief Guest of Inaugural Function, Faculty Development Program, Poornima College of Engineering, Jaipur .
5. "Words of Women in Mathematics in the Time of Corona**,**" part of the short film which brings together statements about how the pandemic has affected personal or professional life as a mathematician launched on May 12th, Maryam Mirzakhani's birthday, a day to celebrate worldwide Women in Maths! It collects words of some 86 women in mathematics from 37 countries telling about their experience in the time of Corona in 25 different languages. This project emanates from a group

of mathematicians, Sonia Mahmoudi (Tohoku U., **Japan**), Claudia Malvenuto (U. di Roma La Sapienza, **Italy**), Sylvie Paycha (U. Potsdam, **Germany**) and Eriko Shinkawa (Tohoku U., **Japan**) after informal conversations about their experiences during the pandemic. https://www.facesofwomeninmathematics.com

6. Selected for **Open Arms Travel Grants program** to receive financial support to attend the International Congress of Mathematicians 2018(ICM-2018), Rio de Janeiro, **Brazil**, Latin America, August 01 – 09, 2018.
7. Selected for waiving off registration fee to attend the World Meeting for Women in Mathematics((WM)^2- 2018), a satellite event of the ICM 2018, Rio de Janeiro, **Brazil**, Latin America, July 31, 2018.
8. Received certificate of Appreciation in International Conference on Advances in Science, Engineering & Technology, International Journal of Advance Research and Innovation and MANU-International Council for Man and Nature, Jan. 29-30, 2016.
9. Received Certificate of Merit in National conference on Recent Trends in Physics, Mathematics and Engineering, Sharda Vilas College Mysore, Feb. 20-21, 2015.
10. Selected to receive full **financial support**, International Centre for Pure and Applied Mathematics (***CIMPA***) to attend CIMPA Research School, Geometrique structures and controle theory, Université Cheikh Anta Diop de Dakar (UCAD) Dakar, **Sénégal(Africa),** May 07-18, 2012.
11. Selected as member of delegate from Uttarakhand to participate in ICM-2010, Hyderabad.
12. Selected to receive full **financial support** to participate in ICM-2010, Hyderabad.
13. Selected to receive full **financial support** to participate in ICWM-2010, Hyderabad.

26.Patent /Design

- **Granted:** Khairul Alam, Yumnam Rohen, **Anita Tomar**. *Processing device for computational fractal analysis* (Design No. 464109-001). The Patent Office, Government of India. Registered June 30, 2025; issued October 13, 2025. No.: 210758. Journal Number: 42/2025. Class: 14-02-Data Processing Equipment as well as Peripheral Apparatus and Devices.